

# Machine Learning

## Dimensionality Reduction

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# Lecture Overview

- 1** Lecture Overview
- 2 Introduction
- 3 Linear Dimensionality Reduction & PCA
- 4 Linear Dimensionality Reduction & Probabilistic PCA
- 5 Non-Linear Dimensionality Reduction
- 6 Summary

# Lecture Overview

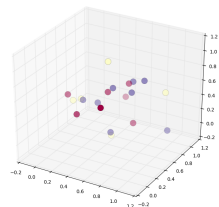
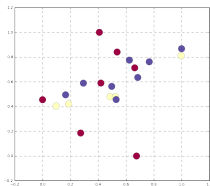
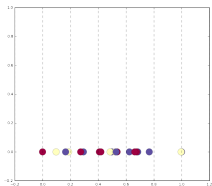
By the end of this lecture you should:

- 1 Understand the problem of the **curse of dimensionality** and further motivations for **dimensionality reduction**
- 2 Know how **Principal Components Analysis** can be used to project data onto a lower-dimensional **subspace**
- 3 Be familiar with **manifold learning** as a means to uncover the non-linear low-dimensional structure in high dimensional data

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- 6 Summary

# The Curse of Dimensionality



- As the dimensionality of our input space increases, the number of instances that we need to 'fill' that space increases

# Specific Motivations

- **Data Visualisation**
- **Data Compression**
- **Denoising**

# Dimensionality Reduction

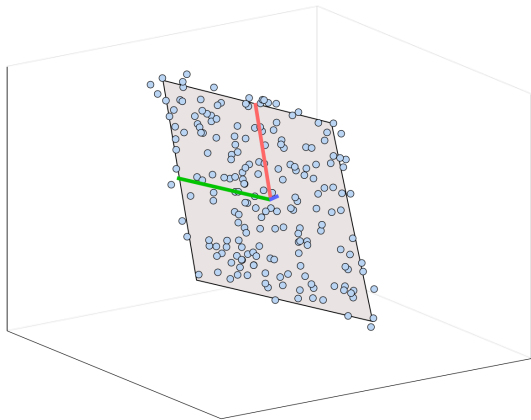
- At the heart of dimensionality reduction techniques is the idea that, although our data is high-dimensional, it actually *lies* on or near a low-dimensional **subspace** or **manifold**
- If we assume our data lies on a subspace, we can use **linear** techniques to obtain a low-dimensional estimation
- If we assume our data lies on a manifold, then we will need to use **non-linear techniques**

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- 5 Non-Linear Dimensionality Reduction
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## Linear Subspaces



The data shown is  $\mathbf{x} \in \mathbb{R}^3$

Along with this data is drawn a hyperplane that passes through both the origin and the data. The hyperplane is referred to as a **subspace** of  $\mathbb{R}^3$

# Linear Subspaces

- The hyperplane is a subset of  $\mathbb{R}^3$ :
  - Each point on the plane is represented by a single point in  $\mathbb{R}^3$ . But...
  - ...We only require  $\mathbb{R}^2$  co-ordinates to describe a single point on the hyperplane
- So if we have a subspace  $\mathbb{R}^d$  within  $\mathbb{R}^m$ , where  $d \ll m$ , then we can reduce the dimensionality of our data

# Principal Components Analysis

- **Principal Component Analysis (PCA)**, is one of the oldest methods for linear dimensionality reduction
- One view of PCA is that of **Projected Variance Maximisation** (other, equivalent views exist):
  - Here the idea is to project our high-dimensional data onto a lower-dimensional subspace (defined up to a rotation)...
  - ...Such that the **sample variance** is maximally preserved
  - This subspace encapsulates the directions along which the data varies the most

## PCA: Setting

- As usual, we assume that our input data consists of  $n$  instances:  $\{\mathbf{x}^{(i)}\}_{i=1}^n$  where:  $\mathbf{x}^{(i)} \in \mathbb{R}^m$
- Our goal is to project this data (linearly) onto a space  $\mathbb{R}^d$ , where  $d < m$ , such that the variance of projected data is maximised
- This space is spanned by the set of basis vectors  $\{\mathbf{u}^{[i]}\}_{i=1}^d$  where:  $\mathbf{u}^{[i]} \in \mathbb{R}^m$
- Since this will not fully define the space in which we are interested, we remove some degeneracy (and ease our mathematical analysis) by requiring **orthonormality**:

$$\mathbf{u}^{[i]} \cdot \mathbf{u}^{[j]} = \delta_{ij}$$

## PCA: Setting

- For each basis vector  $\mathbf{u}^{[j]}$ , each data point  $\mathbf{x}^{(i)}$  is then **projected** onto a scalar value  $\mathbf{u}^{[j]} \cdot \mathbf{x}^{(i)}$
- The mean of the projected data onto this basis vector is  $\mathbf{u}^{[j]} \cdot \bar{\mathbf{x}}$ , where  $\bar{\mathbf{x}}$  is the sample mean:

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)}$$

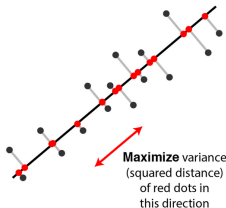
## Projected Variance Maximisation

- The sample variance of the projected data is given by:

$$\frac{1}{n} \sum_{i=1}^n \left( \mathbf{u}^{[j]} \cdot \mathbf{x}^{(i)} - \mathbf{u}^{[j]} \cdot \bar{\mathbf{x}} \right)^2 = \mathbf{u}^{[j]T} \mathbf{S} \mathbf{u}^{[j]}$$

- Where  $\mathbf{S}$  is the **sample covariance matrix** defined by:

$$\mathbf{S} = \frac{1}{n} \sum_{i=1}^n \left( \mathbf{x}^{(i)} - \bar{\mathbf{x}} \right) \left( \mathbf{x}^{(i)} - \bar{\mathbf{x}} \right)^T = \frac{1}{n} \mathbf{X}^T \mathbf{X}$$



## Projected Variance Maximisation

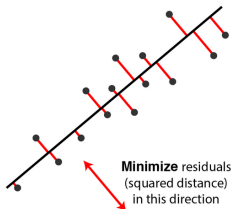
- Here we note that  $\mathbf{X}$  is a **centred design matrix**:

$$\mathbf{X} = \begin{bmatrix} (\mathbf{x}^{(1)} - \bar{\mathbf{x}})^T \\ (\mathbf{x}^{(2)} - \bar{\mathbf{x}})^T \\ \vdots \\ (\mathbf{x}^{(n)} - \bar{\mathbf{x}})^T \end{bmatrix}$$

## Aside: Reconstruction Error Minimisation

- We can also view PCA as a search for the  $d$  dimensional subspace which minimises the **reconstruction error**:

$$\sum_{i=1}^n \left\| (\mathbf{x}^{(i)} - \bar{\mathbf{x}}) - \sum_{j=1}^d \left( \mathbf{u}^{[j]} \cdot (\mathbf{x}^{(i)} - \bar{\mathbf{x}}) \right) \mathbf{u}^{[j]} \right\|_2^2$$





## PCA: Problem Formulation

- We are interested in finding  $\{\mathbf{u}^{[j]}\}_{j=1}^d$  such that the sum of the variance of the projected sample data is maximised
- In other words we wish to solve the following optimisation problem:

$$\operatorname{argmax}_{\{\mathbf{u}^{[j]}\}_{j=1}^d} L \quad \text{where:} \quad L = \sum_{j=1}^d \mathbf{u}^{[j]T} \mathbf{S} \mathbf{u}^{[j]} \quad (1)$$

Subject to orthonormal  $\{\mathbf{u}^{[j]}\}_{j=1}^d$

## Eigendecomposition

- Consider a square symmetric matrix,  $\mathbf{S}$ , with **eigenvalues**  $\{\lambda_i\}_{i=1}^m$ , and associated (orthonormal) **eigenvectors**  $\{\mathbf{q}_i \in \mathbb{R}^m\}_{i=1}^m$ :

$$\mathbf{S}\mathbf{q}_i = \lambda_i\mathbf{q}_i \quad \text{where:} \quad \mathbf{q}_i \cdot \mathbf{q}_j = \delta_{ij}$$

- Then the **eigendecomposition** of  $\mathbf{S}$  is given by:

$$\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

where:

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_m)$$

$$\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_m]$$

## PCA: Greedy Solution

- Let us proceed in a step-wise fashion and attempt to learn each dimension **greedily**
- First, let us search for the direction of highest variance, and write our objective as the following Lagrangian,  $\mathcal{L}$ :

$$\mathcal{L}(\mathbf{u}^{[1]}, \lambda^{[1]}) = \mathbf{u}^{[1]T} \mathbf{S} \mathbf{u}^{[1]} - \lambda^{[1]} (\mathbf{u}^{[1]} \cdot \mathbf{u}^{[1]} - 1)$$

Where  $\lambda^{[1]}$  is a Lagrange multiplier associated with the orthogonality constraint.

- Seeking stationarity wrt  $\mathbf{u}^{[1]}$  gives:

$$\begin{aligned} 2\mathbf{S}\mathbf{u}^{[1]} - 2\lambda^{[1]}\mathbf{u}^{[1]} &= 0 \\ \implies \mathbf{S}\mathbf{u}^{[1]} &= \lambda^{[1]}\mathbf{u}^{[1]} \end{aligned}$$

## PCA: First Basis vector

- Since  $\mathbf{u}^{[1]}$  satisfies the eigenvalue equation this lets us equate it with some eigenvalue  $\mathbf{q}_i$
- But which eigenvalue? Let us calculate the variance of the projected data:

$$\mathbf{u}^{[1]T} \mathbf{S} \mathbf{u}^{[1]} = \lambda^{[1]}$$

- We want to maximise this variance so we select  $\mathbf{u}^{[1]} = \mathbf{q}_1$ , the eigenvector associated with  $\lambda_1$ , the largest eigenvalue. Thus:

$$\mathbf{u}^{[1]} = \mathbf{q}_1$$

$$\lambda^{[1]} = \lambda_1$$

## PCA: Second Basis Vector

- Now let us find another direction,  $\mathbf{u}^{[2]}$ , to further increase the projected variance, such that  $\mathbf{u}^{[2]} \cdot \mathbf{u}^{[2]} = 1$  and  $\mathbf{u}^{[1]} \cdot \mathbf{u}^{[2]} = 0$ :
- We write a Lagrangian,  $\mathcal{L}$ , for this problem as follows:

$$\mathcal{L}(\mathbf{u}^{[2]}, \lambda^{[2]}, \lambda^{[1][2]}) = \mathbf{u}^{[2]T} \mathbf{S} \mathbf{u}^{[2]} - \lambda^{[2]} (\mathbf{u}^{[2]} \cdot \mathbf{u}^{[2]} - 1) - \lambda^{[1][2]} (\mathbf{u}^{[2]} \cdot \mathbf{u}^{[1]} - 0)$$

Where  $\lambda^{[2]}$  and  $\lambda^{[1][2]}$  are Lagrange multipliers.

- Seeking stationarity wrt  $\mathbf{u}^{[2]}$  gives:

$$2\mathbf{S}\mathbf{u}^{[2]} - 2\lambda^{[2]}\mathbf{u}^{[2]} - \lambda^{[1][2]}\mathbf{u}^{[1]} = 0 \quad (2)$$

## PCA: Second Basis Vector

- Left multiply equation (2) by  $\mathbf{u}^{[1]T}$  gives:

$$\implies 2\mathbf{u}^{[1]T}\mathbf{S}\mathbf{u}^{[2]} - 2\lambda^{[2]}\mathbf{u}^{[1]} \cdot \mathbf{u}^{[2]} - \lambda^{[1][2]}\mathbf{u}^{[1]} \cdot \mathbf{u}^{[1]} = 0$$

$$\implies 2\mathbf{u}^{[2]T}\mathbf{S}\mathbf{u}^{[1]} - \lambda^{[1][2]} = 0$$

$$\implies 2\lambda^{[1]}\mathbf{u}^{[2]} \cdot \mathbf{u}^{[1]} - \lambda^{[1][2]} = 0$$

$$\implies \lambda^{[1][2]} = 0$$

- Therefore:

$$\mathbf{S}\mathbf{u}^{[2]} = \lambda^{[2]}\mathbf{u}^{[2]}$$

## PCA: Second Basis vector

- Since  $\mathbf{u}^{[2]}$  satisfies the eigenvalue equation this lets us equate it with some eigenvalue  $\mathbf{q}_i$
- But which eigenvalue? Let us calculate the variance of the projected data:

$$\mathbf{u}^{[2]T} \mathbf{S} \mathbf{u}^{[2]} = \lambda^{[2]}$$

- We want to maximise this variance so we select  $\mathbf{u}^{[2]} = \mathbf{q}_2$ , the eigenvector associated with  $\lambda_2$ , the largest remaining eigenvalue. Thus:

$$\mathbf{u}^{[2]} = \mathbf{q}_2$$

$$\lambda^{[2]} = \lambda_2$$

## PCA: Subsequent Basis Vectors

- The solution proceeds in a similar step-wise fashion, to give:

$$\left\{ \mathbf{u}^{[j]*} = \mathbf{q}_j \right\}_{j=1}^d$$

- And this implies a total projected variance of

$$L^* = \sum_{j=1}^d \mathbf{q}_j^T \mathbf{S} \mathbf{q}_j = \sum_{j=1}^d \lambda_j$$



## PCA: Non-Uniqueness

- Note that we have made a number of choices in this derivation...
  - Orthogonality of basis vectors
  - Normality of different basis vectors
  - Orientation of the first basis vector
- Other choices are possible which would also yield similar solutions
- This suggests that we should check if a particular similar solution is globally optimal

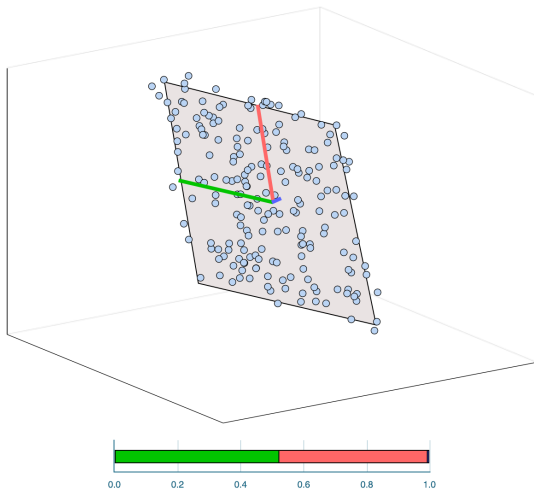
## PCA: Non-Convexity

- The trouble is that our original problem is **non-convex**
- However we can show that the greedy local optima we have investigated give rise to globally optimal points
- It turns out that PCA is a rare example of a non-convex optimisation problem which we can solve globally!

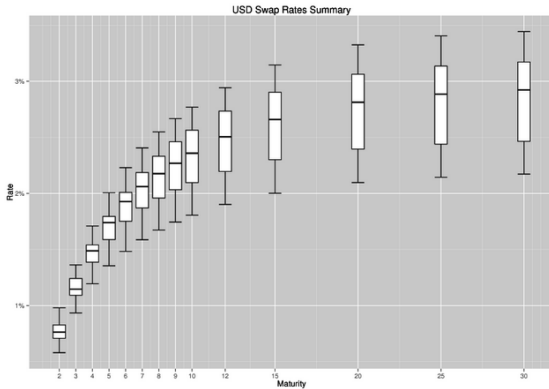
## PCA: Examining the Eigenvalues

- Our PCA solution allows us to estimate the subspace as the space spanned by the first  $d$  eigenvectors of the sample covariance matrix  $\mathbf{S}$  ordered by size of eigenvalue
- If we refer to the dimensionality of the input space as the **ambient dimensionality** of the data, then the **intrinsic dimensionality** is the dimensionality of the subspace upon which the data is assumed to lie

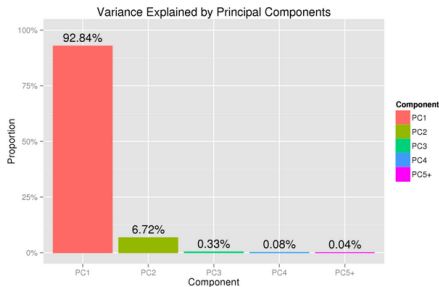
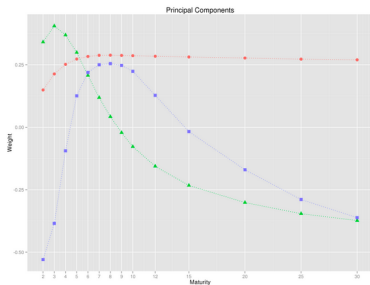
# Examining the Eigenvalues



## Application: Swaps Curve



## Application: Swaps Curve



■ ‘Shift’, ‘Tilt’, ‘Curvature’ effects dominate

# Pattern Stability

- PCA is intuitively plausible, but can we be sure that our training sample has allowed us to discern a reliable subspace?
- At least two possible responses to this, which flow from different ML paradigms:

## Pattern Stability

### ■ Generative Modelling:

- *Probabilistic PCA* - Seeks a Gaussian latent variable model
- Learn parameters using MLE or Bayesian treatment

### ■ PAC Approach:

- Seeks to bound generalisation error resulting from projection:  
$$\mathbb{E}_{\mathcal{D}} \left[ \mathbf{x} - \sum_{j=1}^d (\mathbf{u}^{[j]} \cdot \mathbf{x}) \mathbf{u}^{[j]} \right]^2, \text{ with high probability}$$
- Bound contains terms in sample residual eigenspectrum and in complexity of data space
- Indicates low generalisation error if subspace captures a high proportion of the variance in a dimensionality small compared to training data size



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# Probabilistic PCA

- Let's investigate the probabilistic generalisation of PCA
- Here it's valuable to recall the probabilistic generalisation of the  $k$ -means model to the Mixture of Gaussians model for clustering
- Can we build a Latent Variable version of PCA?

## PPCA: Setting

- As before, our input data consists of  $n$  instances,  $\{\mathbf{x}^{(i)} \in \mathbb{R}^m\}_{i=1}^n$ , with sample mean  $\bar{\mathbf{x}}$ , and associated sample covariance matrix,  $\mathbf{S} = \frac{1}{n} \mathbf{X}^T \mathbf{X}$
- And as before, we seek some  $d$ -dimensional principal component subspace, where  $d < m$

## PPCA: Model

- Each point has an unknown, latent, variable,  $\mathbf{z} \in \mathbb{R}^d$  associated with it, corresponding to its position in the principal component subspace. This variable is the outcome of a Gaussian random variable,  $\mathcal{Z}$ , such that:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

- Contingent on the principal component subspace variable, each  $\mathbf{x}$  is the outcome of a Gaussian random variable,  $\mathcal{X}$ , such that:

$$\mathbf{x}|\mathbf{z} \sim \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I}_m)$$

where:  $\mathbf{W} \in \mathbb{R}^{m \times d}$ , which defines the directions of the principal subspace,  $\boldsymbol{\mu} \in \mathbb{R}^m$ ,  $\sigma \in \mathbb{R}^+$

## PPCA: Model

- We can view the model from a generative standpoint:
  - First a value is drawn for the latent variable,  $\mathbf{z}$
  - Then the observed variable is sampled, conditional on this latent variable,  $\mathbf{x}|\mathbf{z}$
  - And residual noise is captured by  $\epsilon$ , which is an outcome of an  $m$ -dimensional random variable,  $\epsilon$ , distributed like a zero-mean Gaussian:

$$\mathbf{x} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon} \quad (3)$$

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_m)$$

where  $\mathbf{z}$  and  $\epsilon$  are uncorrelated

## PPCA: Model

- Recall the properties of the **marginal & conditional distributions** associated with **Linear Gaussian Models** that we encountered in the *Probability Lecture*:
  - Given a marginal distribution for  $\tilde{\mathbf{x}}$  and a conditional Gaussian distribution for  $\tilde{\mathbf{y}}$  given  $\tilde{\mathbf{x}}$ :

$$\tilde{\mathbf{x}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$$

$$\tilde{\mathbf{y}}|\tilde{\mathbf{x}} \sim \mathcal{N}(\mathbf{A}\tilde{\mathbf{x}} + \mathbf{b}, \mathbf{L}^{-1})$$

Where:  $\tilde{\mathbf{x}} \in \mathbb{R}^n$ ,  $\tilde{\mathbf{y}} \in \mathbb{R}^m$ ,  $\boldsymbol{\mu} \in \mathbb{R}^n$ ,  $\boldsymbol{\Lambda} \in \mathbb{R}^{n \times n}$ ,  $\mathbf{A} \in \mathbb{R}^{m \times n}$ ,  $\mathbf{b} \in \mathbb{R}^m$ ,  
 $\mathbf{L} \in \mathbb{R}^{m \times m}$

Then:

$$\tilde{\mathbf{y}} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^T)$$

$$\tilde{\mathbf{x}}|\tilde{\mathbf{y}} \sim \mathcal{N}(\boldsymbol{\Sigma}[\mathbf{A}^T\mathbf{L}(\tilde{\mathbf{y}} - \mathbf{b}) + \boldsymbol{\Lambda}\boldsymbol{\mu}], \boldsymbol{\Sigma})$$

Where:  $\boldsymbol{\Sigma} = (\boldsymbol{\Lambda} + \mathbf{A}^T\mathbf{L}\mathbf{A})^{-1}$

## PPCA: Model

- From this, we can see that  $\mathbf{x}$  is drawn from a Gaussian distribution with:

- Mean:

$$\begin{aligned}\mathbb{E}[\mathbf{X}] &= \mathbf{W}\mathbb{E}[\mathbf{Z}] + \boldsymbol{\mu} + \mathbb{E}[\boldsymbol{\epsilon}] \\ &= \boldsymbol{\mu}\end{aligned}$$

- Covariance:

$$\begin{aligned}\mathbf{C} &= \mathbb{E}[(\mathbf{W}\mathbf{Z} + \boldsymbol{\mu} + \boldsymbol{\epsilon} - \mathbb{E}[\mathbf{X}])(\mathbf{W}\mathbf{Z} + \boldsymbol{\mu} + \boldsymbol{\epsilon} - \mathbb{E}[\mathbf{X}])^T] \\ &= \mathbb{E}[(\mathbf{W}\mathbf{Z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{Z} + \boldsymbol{\epsilon})^T] \\ &= \mathbb{E}[\mathbf{W}\mathbf{Z}\mathbf{Z}^T\mathbf{W}^T] + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] + \mathbb{E}[\mathbf{W}\mathbf{Z}\boldsymbol{\epsilon}^T] + \mathbb{E}[\boldsymbol{\epsilon}\mathbf{Z}^T\mathbf{W}^T] \\ &= \mathbf{W}\mathbb{E}[\mathbf{Z}\mathbf{Z}^T]\mathbf{W}^T + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\ &= \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}_m\end{aligned}$$

- Thus:

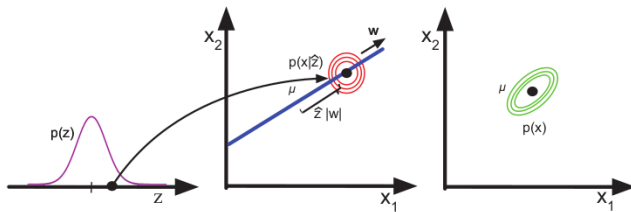
$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$$

## PPCA: Interpretation

- We can interpret  $p_{\mathbf{x}}(\mathbf{x})$  as a density defined by taking an isotropic Gaussian ‘spray can’...
- ...Then moving across the principal subspace, spraying Gaussian ink with density determined by  $\sigma^2$ ...
- ...And weighted by the prior distribution,  $p_{\mathbf{z}}(\mathbf{z})$
- This results in an ink density which has a pancake shaped distribution, which represents  $p_{\mathbf{x}}(\mathbf{x})$



# PPCA: Interpretation



**Figure 12.1** Illustration of the PPCA generative process, where we have  $L = 1$  latent dimension generating  $D = 2$  observed dimensions. Based on Figure 12.9 of (Bishop 2006b).

## PPCA: Rotational Invariance

- Let  $\mathbf{R}$  be an orthogonal (rotation) matrix (for which  $\mathbf{R}^T \mathbf{R} = \mathbf{I}$ )
- Now apply this rotation to the latent space coordinate matrix,  $\mathbf{W}$ :

$$\begin{aligned}\tilde{\mathbf{W}} &= \mathbf{W}\mathbf{R} \\ \implies \tilde{\mathbf{W}}\tilde{\mathbf{W}}^T &= \mathbf{W}\mathbf{R}\mathbf{R}^T\mathbf{W}^T \\ &= \mathbf{W}\mathbf{W}^T\end{aligned}$$

- Thus  $p_{\mathcal{X}}(\mathbf{x})$  is as well characterised by any  $\tilde{\mathbf{W}}$  as it is by  $\mathbf{W}$
- This is the analogue of the non-uniqueness which we encountered in PCA

## PPCA: Learning Problem

- Now that we have defined our generative model we need to learn its parameters:  $\mu$ ,  $\mathbf{W}$ ,  $\sigma^2$
- Let's use a Maximum Likelihood approach:

## PPCA: Log Likelihood

$$\begin{aligned}\ln \mathbb{P} \left( \{\mathbf{x}^{(i)}\}_{i=1}^n \right) &= \sum_{i=1}^n \ln p_{\mathcal{X}}(\mathbf{x}^{(i)}; \mathbf{W}, \boldsymbol{\mu}, \sigma^2) \\ &= -\frac{nm}{2} \ln(2\pi) - \frac{n}{2} \ln |\mathbf{C}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}^{(i)} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu})\end{aligned}$$

## PPCA: Optimisation

- Tipping & Bishop ('99) demonstrate the following closed form solutions which flow from the optimisation of this function:



$$\mu_{\text{MLE}} = \bar{\mathbf{x}}$$



$$\mathbf{W}_{\text{MLE}} = \mathbf{Q}(\mathbf{\Lambda} - \sigma^2 \mathbf{I})^{1/2} \mathbf{R}$$

Here:

$\mathbf{Q}$  is the  $m \times d$  matrix whose columns are given by the leading  $d$  eigenvectors of the covariance matrix  $\mathbf{S}$

$\mathbf{\Lambda}$  is the diagonal matrix of the  $d$  leading eigenvalues associated with these eigenvectors

$\mathbf{R}$  is an arbitrary orthogonal matrix



$$\sigma_{\text{MLE}}^2 = \frac{1}{m-d} \sum_{i=d+1}^m \lambda_i$$

## PPCA: Interpretation of Solution

- We can set  $\mathbf{R} = \mathbf{I}$  without loss of generality, in which case the columns of  $\mathbf{W}$  are the principal component eigenvectors scaled by the square root of the variance parameters  $(\lambda_i - \sigma^2)^{1/2}$
- So the variance of  $p_{\mathcal{X}}(\mathbf{x})$  in the  $\mathbf{q}_i$  direction is given by:

$$\begin{aligned}\mathbf{q}_i^T \mathbf{C} \mathbf{q}_i &= \mathbf{q}_i^T \mathbf{W} \mathbf{W}^T \mathbf{q}_i + \sigma^2 \mathbf{q}_i^T \mathbf{q}_i \\ &= \lambda_i - \sigma^2 + \sigma^2 \\ &= \lambda_i\end{aligned}$$

- So this model captures the variance of the data in the direction of the principal axes
- While  $\sigma_{\text{MLE}}^2$  is the average variance associated with the discarded dimensions

## PPCA: Connection with PCA

- Recall the distributional forms for  $\mathbf{z}$ , and  $\mathbf{x}|\mathbf{z}$ :

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

$$\mathbf{x}|\mathbf{z} \sim \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I}_n)$$

- Then using the conditional distribution property for Linear Gaussian Models once again:

$$\mathbf{z}|\mathbf{x} \sim \mathcal{N}(\mathbf{M}^{-1}\mathbf{W}_{\text{MLE}}^T(\mathbf{x} - \boldsymbol{\mu}), \sigma^2\mathbf{M}^{-1})$$

$$\text{Where: } \mathbf{M} = \mathbf{W}_{\text{MLE}}^T \mathbf{W}_{\text{MLE}} + \sigma^2\mathbf{I}$$

- This is the particular distribution of  $\mathbf{z}$  given a point  $\mathbf{x}$  in the input data space.

## PPCA: Connection with PCA

- The expectation of  $\mathcal{Z}|\mathbf{x}$  gives a summary of the point  $\mathbf{x}$  in latent space:

$$\mathbb{E}[\mathcal{Z}|\mathbf{x}] = \mathbf{M}^{-1}\mathbf{W}_{\text{MLE}}^T(\mathbf{x} - \boldsymbol{\mu})$$



## PPCA: Connection with PCA

- And this latent variable point projects back to a point in the input data space the expectation of which is given by:

$$\begin{aligned} & \mathbf{W}_{\text{MLE}} \mathbb{E}[\mathbf{z}|\mathbf{x}] + \boldsymbol{\mu} \\ &= \mathbf{W}_{\text{MLE}} \mathbf{M}^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \end{aligned}$$

## PPCA: Connection with PCA

- And this latent variable point projects back to a point in the input data space the expectation of which is given by:

$$\begin{aligned}\mathbf{W}_{\text{MLE}} \mathbb{E}[\mathbf{z}|\mathbf{x}] + \boldsymbol{\mu} \\&= \mathbf{W}_{\text{MLE}} \mathbf{M}^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\&= \mathbf{W}_{\text{MLE}} (\mathbf{W}_{\text{MLE}}^T \mathbf{W}_{\text{MLE}} + \sigma^2 \mathbf{I})^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu}\end{aligned}$$

## PPCA: Connection with PCA

- And this latent variable point projects back to a point in the input data space the expectation of which is given by:

$$\begin{aligned} & \mathbf{W}_{\text{MLE}} \mathbb{E}[\mathbf{z}|\mathbf{x}] + \boldsymbol{\mu} \\ &= \mathbf{W}_{\text{MLE}} \mathbf{M}^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\ &= \mathbf{W}_{\text{MLE}} \left( \mathbf{W}_{\text{MLE}}^T \mathbf{W}_{\text{MLE}} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\ &= \mathbf{W}_{\text{MLE}} \left( (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \mathbf{Q}^T \mathbf{Q} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \end{aligned}$$

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 &= \mathbf{W}_{\text{MLE}} (\mathbf{W}_{\text{MLE}}^T \mathbf{W}_{\text{MLE}} + \sigma^2 \mathbf{I})^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{W}_{\text{MLE}} \left( (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \mathbf{Q}^T \mathbf{Q} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{W}_{\text{MLE}} \boldsymbol{\Lambda}^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{Q} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \boldsymbol{\Lambda}^{-1} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \mathbf{Q}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu}
 \end{aligned}$$

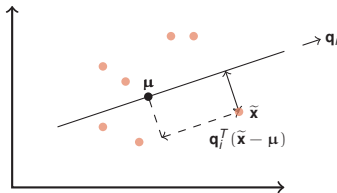
## PPCA: Connection with PCA

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 &= \mathbf{W}_{\text{MLE}} \left( (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \mathbf{Q}^T \mathbf{Q} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{W}_{\text{MLE}} \boldsymbol{\Lambda}^{-1} \mathbf{W}_{\text{MLE}}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{Q} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \boldsymbol{\Lambda}^{-1} (\boldsymbol{\Lambda} - \sigma^2 \mathbf{I})^{\frac{1}{2}} \mathbf{Q}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu} \\
 &= \mathbf{Q} \text{diag} \left( \frac{\lambda_1 - \sigma^2}{\lambda_1}, \dots, \frac{\lambda_d - \sigma^2}{\lambda_d} \right) \mathbf{Q}^T (\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu}
 \end{aligned}$$

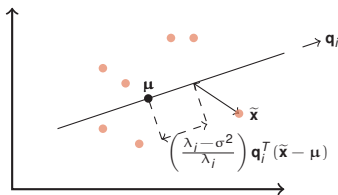
## PPCA: Connection with PCA

- Now, as  $\sigma^2 \rightarrow 0$ , then  $(\mathbf{W}_{\text{MLE}} \mathbb{E}[\mathbf{Z}|\mathbf{x}] + \boldsymbol{\mu}) \rightarrow (\mathbf{Q}\mathbf{Q}^T(\mathbf{x} - \boldsymbol{\mu}) + \boldsymbol{\mu})$
- Thus each data point is approximated by a mapping into a linear subspace defined by the eigenvectors of  $\mathbf{S}$ , given by  $\mathbf{Q}$ , such that each point is orthogonally projected into this subspace
- ...Just as in PCA:



## PPCA: Connection with PCA

- But for  $\sigma^2 > 0$ , each projection is scaled by  $\frac{\lambda_i - \sigma^2}{\lambda_i} < 1$ , thus the projection is shrunk towards  $\mu$ :

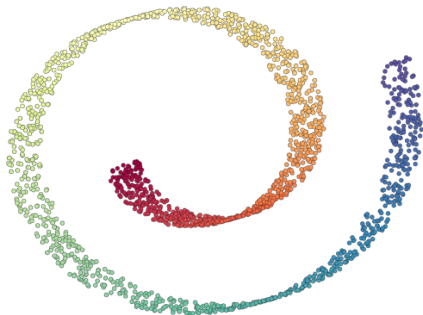
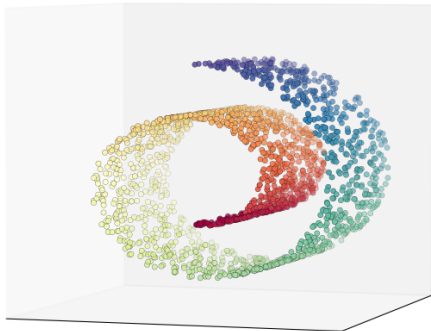




## PPCA: Setting $d$

- Just as in the Mixture of Gaussians model for clustering, we can use our generative model to select  $d$  in a principled way:
- Evaluate the likelihood of data on a validation set for various settings of  $d$  and select the one which gives rise to the maximal one
- A Bayesian treatment of PPCA or the maximisation of the PAC bound offer different resolutions to this problem

## Where PCA Fails

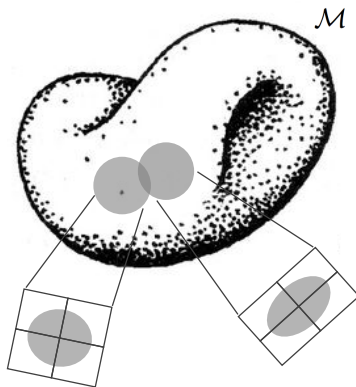


- One of the restrictions of PCA is that it assumes the data lies on or near a linear subspace
- What happens if that assumption fails?

# Lecture Overview

- 1 Lecture Overview
- 2 Introduction
- 3 Linear Dimensionality Reduction & PCA
- 4 Linear Dimensionality Reduction & Probabilistic PCA
- 5 Non-Linear Dimensionality Reduction**
- 6 Summary

# From Subspaces to (Sub)Manifolds



# Manifold Learning Techniques

- Kernel PCA
- Isomap
- Locally Linear Embedding (LLE)
- Autoencoder
- ...

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# Summary

- 1 Dimensionality Reduction** falls into two broad categories depending on whether the low-dimensional space we are mapping to is linear or non-linear
- 2 Principal Component Analysis** is a linear technique for reducing the dimensionality of the data by projecting it onto the maximum covariance subspace
- 3** Manifold learning techniques allow for the linear assumption inherent in PCA to be relaxed